The Invariance Principle — problem 1.38

A. Engel — Problem-Solving Strategies · 1998

The integers $1, \ldots, n$ are arranged in order. In one step you may take any four integers and interchange the first with the fourth and the second with the third. Prove that, if $n(n-1)/2$ is even, then by means of such steps you may reach the arrangement $n, n-1, \ldots, 1$. But if $n(n-1)/2$ is odd, you cannot reach this arrangement.